

Webots Learning Log - June 30, 2026

Main: How Objects Become Real in Webots

Today's big takeaway was understanding the three layers that make up an object in Webots

- **Shape** (under children): The visual layer such as mesh, texture, color, appearance. This is what you see, but it's purely for design. Without anything else, it's basically a hologram.
- **boundingObject**: The collision layer. It's an invisible reference that the physics engine uses to detect contacts. This is what makes an object solid. When I created a floor without a boundingObject, things placed on the floor would fall right through it.
- **Physics**: The dynamics layer that gives the object mass and makes it respond to gravity and forces. Without it the object is static cannot be moved but still collidable if it has a boundingObject.

The relationship between them:

Configuration	Behavior
Shape only	Visible but not physical
Shape + boundingObject	Static obstacle. Things collide with it, but it doesn't move (walls, floors)
Shape + boundingObject + Physics	Fully dynamic

DEF/USE vs PROTO

- **DEF/USE** is a reference system inherited from VRML. Once I define a node once (DEF BALL_GEOMETRY Shape {my pick}), I can reference it elsewhere by using (USE BALL_GEOMETRY). All instances share the exact same data. If I change one, change all. Good for memory efficiency and consistency.
- **PROTO** is what I actually want for reusable templates. It works like a class/prefab where I define fields (radius, color, etc.) and each instance can have its own values.

Gotchas I Ran Into

- **Blank simulation view**: The controller had a variable name mismatch (timestep vs TIME_STEP), which crashed the controller silently. Always check the Console panel.

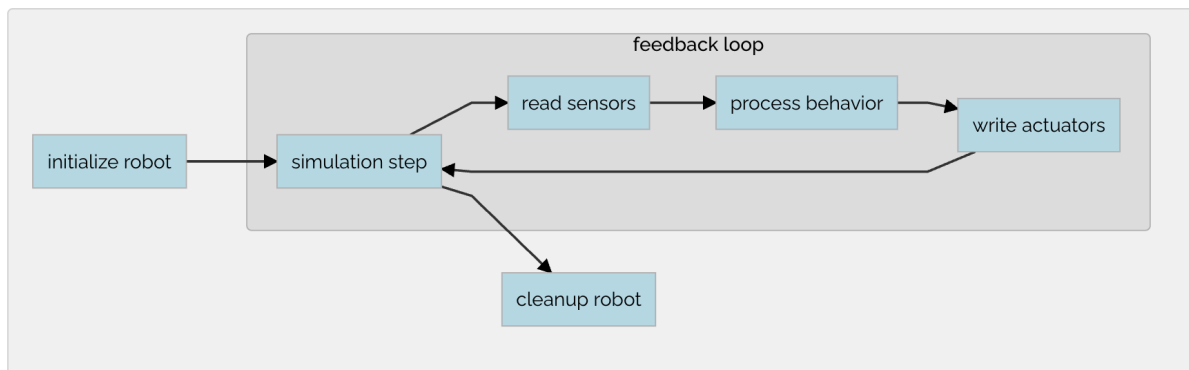
- **Objects falling through the floor:** Both the falling object *and* the surface need valid boundingObject nodes. Collision requires two participants.
- **Applied force but object didn't move:** No Physics node = static object. Also, geometry matters that a cylinder standing upright won't roll like a sphere. Orientation affects behavior.
- **Wireframe overlay confusion:** When you select a boundingObject in the scene tree, Webots highlights it as a blue wireframe in the 3D view.

Building Walls

Walls should be **static** (boundingObject but no Physics). They need to be collidable so objects bounce off them, but they shouldn't move. A simple wall is just a Solid with a Box geometry and matching boundingObject.

Geometry directly affects physics behavior. Same force, same mass, same physics settings but a sphere rolls freely in any direction while a cylinder only rolls perpendicular to its length. Shape isn't just visual. It fundamentally changes how objects interact with the world.

Control System, Feedback Loop



UML state machine of a simple feedback loop

The E-Puck collision avoidance example was a good intro to how feedback loops work in robotics.

The basic cycle: initialize the robot and its sensors → run a simulation step → read what the sensors see → decide what to do → send commands to the motors → repeat.

In this case, the E-Puck has 8 distance sensors around its body. Each loop iteration, it checks if anything is too close. If a sensor on the right side picks up an obstacle past a

certain threshold, it speeds up the left motor to turn away. Same logic mirrored for the other side. Simple rule, but it's enough to navigate around an arena without crashing.

That sense → decide → act loop is pretty much the foundation of any reactive robot controller.